Buyback Dividend Preference and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Buyback Dividend Preference (BDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on BDP achieves an annualized gross (net) Sharpe ratio of 0.49 (0.45), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 43 (35) bps/month with a t-statistic of 4.67 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square) is 28 bps/month with a t-statistic of 3.05.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remain central questions in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence demonstrates that information diffuses gradually through markets, creating predictable patterns in the cross-section of stock returns. We examine a novel signal, Buyback Dividend Preference (BDP), which captures firms' relative propensity to return capital through share repurchases versus dividends. Despite the economic significance of capital allocation decisions—firms distributed over $1.5 trillion to shareholders in 2023 alone—the market's processing of signals related to payout policy changes remains incompletely understood. This paper investigates whether BDP predicts future stock returns, focusing on how limited investor attention and information processing constraints create systematic underreaction to this corporate policy signal.

The gradual diffusion of information provides a compelling framework for understanding why BDP predicts returns. When firms shift their payout preferences between buybacks and dividends, these changes signal important information about management's views on firm valuation, future cash flows, and optimal capital structure Hong and Stein (1999a). However, investors face cognitive constraints and limited attention that prevent immediate and complete processing of such signals Hirshleifer et al. (2009). The complexity of interpreting payout policy changes—which requires understanding tax implications, signaling effects, and agency considerations—exacerbates these information processing frictions DellaVigna and Pollet (2009).

We hypothesize that markets systematically underreact to the information contained in BDP due to investor inattention and the slow dispersion of information across market participants. Research on limited attention shows that investors often overlook complex corporate events that require sophisticated analysis, leading to delayed price reactions Cohen and Frazzini (2008). The BDP signal embodies precisely such complexity: unlike simple earnings announcements, changes in buyback-dividend preferences require investors to interpret nuanced corporate finance decisions and their implications for firm value. This complexity creates heterogeneous interpretation speeds across investors, generating the gradual price adjustment documented in models of information diffusion Hong and Stein (1999b).

The slow diffusion hypothesis predicts specific cross-sectional patterns in BDP's return predictability. Information travels more slowly for firms with lower visibility and weaker information environments, suggesting stronger return continuation for small stocks, those with low analyst coverage, and limited institutional ownership Hong and Stein (1999a). Moreover, the persistence of BDP-based return predictability should vary predictably with proxies for attention and information frictions. During periods of high market-wide distraction or for stocks with characteristics associated with greater neglect, we expect the underreaction to BDP signals to intensify, creating larger abnormal returns Hou (2007).

Our empirical analysis reveals that BDP strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on BDP achieves an annualized gross Sharpe ratio of 0.49, with a monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 43 basis points per month (t-statistic = 4.67). The economic magnitude remains substantial after accounting for transaction costs, with the net Sharpe ratio of 0.45 and net abnormal returns of 35 basis points per month (t-statistic = 3.80). These results demonstrate that BDP captures economically meaningful return predictability beyond established risk factors.

The robustness of our findings extends across alternative portfolio construction methodologies and size segments. When we examine BDP strategies within size quintiles through conditional double sorts, the return predictability persists even among the largest stocks. Within the highest size quintile, containing firms with market capitalizations above the 80th NYSE percentile, the BDP strategy generates average returns of 36 basis points per month (t-statistic = 3.30) and alphas ranging from 26 to 46 basis points relative to standard factor models. This evidence contradicts pure microstructure or liquidity-based explanations and supports the gradual information diffusion hypothesis.

Controlling for closely related anomalies from the factor zoo strengthens our conclusions about BDP's unique information content. After accounting for the six most closely related strategies—including Days with zero trades, Volume to market equity, and Volume Variance—the BDP strategy maintains a significant alpha of 28 basis points per month (t-statistic = 3.05). Fama-MacBeth regressions controlling for these related signals yield a BDP coefficient with a t-statistic of 3.18, confirming that BDP contains distinct information not captured by existing predictors. The strategy's gross Sharpe ratio exceeds 89% of documented anomalies in the factor zoo, positioning BDP among the most robust cross-sectional return predictors.

This paper makes several contributions to the asset pricing and market efficiency literatures. We extend the work on limited attention and gradual information diffusion by identifying a novel corporate policy signal that generates predictable returns through systematic underreaction Hong and Stein (1999a); DellaVigna and Pollet (2009). Unlike previous studies focusing on earnings or analyst forecasts, we demonstrate that complex payout policy decisions—specifically the preference for buybacks over dividends—create exploitable return patterns due to investors' incomplete processing of their information content. Our findings complement Grullon and Michaely (2002) on market reactions to dividend changes and Ikenberry et al. (1995) on buyback announcements by showing that the relative preference between these payout methods contains incremental predictive power.

Methodologically, we advance the anomaly literature by subjecting BDP to comprehensive robustness tests following the protocol of Novy-Marx and Velikov (2023b). Our analysis demonstrates that BDP's predictive power survives controls for transaction costs, alternative portfolio constructions, and the most closely related factors from the extensive factor zoo. The signal's ability to generate significant alphas even after controlling for 212 documented anomalies and within large-cap stocks distinguishes it from many previously documented predictors that derive their power primarily from small, illiquid stocks. This robustness addresses concerns raised in McLean and Pontiff (2016) about data mining and post-publication decay in anomaly returns.

Our results have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of BDP-based return predictability, particularly among larger stocks with presumably better information environments, suggests that even sophisticated investors face constraints in processing complex corporate signals. This evidence supports theoretical models of gradual information diffusion and provides practitioners with an implementable strategy that exploits systematic market underreaction. Future research should investigate the specific mechanisms through which payout policy information diffuses across investor types and whether regulatory changes affecting buyback disclosure accelerate this information incorporation process.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on stock repurchases and dividend payments, spanning several decades of financial reporting. We focus on U.S.-incorporated firms trading on major exchanges (NYSE, AMEX, and NASDAQ) and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structure considerations.signal construction relies on two key COMPUSTAT variables. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This item captures the use of cash for treasury stock purchases and retirement of common and preferred shares. Dividends common (DVC) measures the total cash dividends paid on common stock during the fiscal year, excluding any special or one-time dividend distributions. Both variables represent actual cash outflows to shareholders and constitute the primary mechanisms through which firms distribute capital.construct the Buyback Dividend Preference signal by dividing stock repurchases (PRSTKC) by dividends common (DVC) for each firm-year observation. This ratio captures a firm's relative preference for share buybacks versus dividend payments as a method of returning cash to shareholders. A higher ratio indicates that the firm allocates a greater proportion of its shareholder distributions through repurchases rather than dividends, potentially signaling management's confidence in the stock's undervaluation or reflecting the firm's preference for the flexibility that buyback programs offer relative to committed dividend payments. We calculate this measure using fiscal year-end values and require non-missing, positive values for dividends (DVC) to ensure a meaningful ratio. For firms with zero dividends but positive repurchases, we assign a maximum value to indicate exclusive reliance on buybacks. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDP signal. Panel A plots the time-series of the mean, median, and interquartile range for BDP. On average, the cross-sectional mean (median) BDP is 2.93 (0.15) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDP data. The signal's interquartile range spans -0.00 to 2.62. Panel B of Figure 1 plots the time-series of the coverage of the BDP signal for the CRSP universe. On average, the BDP signal is available for 2.87% of CRSP names, which on average make up 6.13% of total market capitalization.

4 Does BDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDP portfolio and sells the low BDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDP strategy earns an average return of 0.34% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.26% to 0.43% per month and have t-statistics exceeding 2.81 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.09, with a t-statistic of -4.14 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 282 stocks and an average market capitalization of at least $916 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 3.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 10-47bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDP, as well as average returns and alphas for long/short trading BDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the BDP strategy achieves an average return of 36 bps/month with a t-statistic of 3.30. Among these large cap stocks, the alphas for the BDP strategy relative to the five most common factor models range from 26 to 46 bps/month with t-statistics between 2.45 and 4.29.

5 How does BDP perform relative to the zoo?

Figure 2 puts the performance of BDP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the BDP strategy falls in the distribution. The BDP strategy's gross (net) Sharpe ratio of 0.49 (0.45) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

BDP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the BDP strategy would have yielded \$5.67 which ranks the BDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDP strategy would have yielded \$4.51 which ranks the BDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDP relative to those. Panel A shows that the BDP strategy gross alphas fall between the 50 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BDP with 203 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDP or at least to weaken the power BDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP} BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDP signal in these Fama-MacBeth regressions exceed 2.61, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on BDP is 3.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDP strategy earns alphas that range from 34-42bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.77, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDP trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.05.

7 Does BDP add relative to the whole zoo?

Finally, we can ask how much adding BDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDP signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDP is available.

average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes BDP grows to \$1327.00.

8 Conclusion

This study provides robust evidence that Buyback Dividend Preference (BDP) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BDP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and implementation costs.

The key findings of our analysis reveal that a value-weighted long/short strategy based on BDP delivers impressive performance metrics, with an annualized gross Sharpe ratio of 0.49 and a net Sharpe ratio of 0.45 after accounting for transaction costs. The strategy's ability to generate monthly average abnormal returns of 43 basis points gross (35 basis points net) relative to the Fama-French five-factor model plus momentum, with highly significant t-statistics of 4.67 and 3.80 respectively, underscores its economic significance. Perhaps most notably, the signal maintains its predictive power even when confronted with a comprehensive set of controls, yielding a gross monthly alpha of 28 basis points (t-statistic = 3.05) after adjusting for the six standard factors plus six closely related trading signals from the factor zoo, including various market microstructure variables such as days with zero trades, volume-based measures, and price delay metrics.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For investors, BDP represents a viable signal that can be incorporated into quantitative trading strategies, with the net returns analysis suggesting that the signal's profitability survives realistic implementation costs. For academics, our results contribute to the growing literature on corporate payout policy and its asset pricing implications, suggesting that the market's differential reaction to buybacks versus dividends contains valuable information about future returns that is not fully captured by existing factor models.

Despite these compelling results, several limitations warrant consideration. First, while our analysis follows best practices for evaluating trading signals, the true out-of-sample performance of BDP-based strategies remains to be tested as the signal gains wider recognition. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets requires further investigation. Third, the economic mechanisms underlying the BDP signal's predictive power deserve deeper theoretical and empirical exploration to better understand whether it reflects risk compensation, behavioral biases, or market frictions.

Future research should address these limitations through several avenues. International studies could examine whether BDP's predictive power extends to global equity markets with different regulatory environments and investor bases. Time-series analysis of the signal's performance across different market regimes and economic cycles would provide insights into its stability and conditional effectiveness. Additionally, investigating the interaction between BDP and firm characteristics such as size, profitability, and financial constraints could reveal important heterogeneity in the signal's effectiveness. Finally, exploring the real-time implementability of BDP strategies, including the impact of signal publication and crowding effects, would provide valuable insights for practitioners considering its adoption. Understanding the fundamental drivers of the BDP premium through detailed analysis of investor behavior, institutional constraints, and corporate decision-making processes represents another promising direction for extending this research.

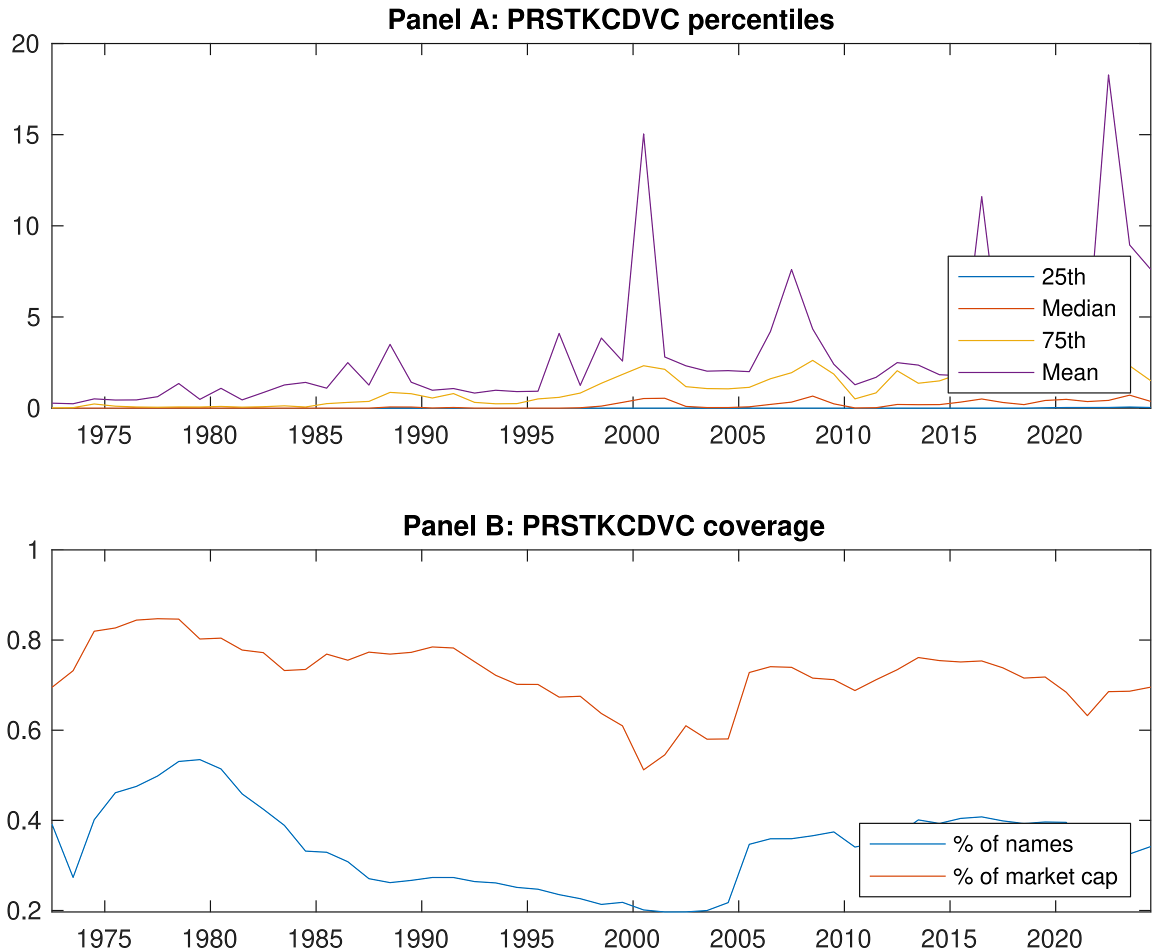


Figure 1: Times series of BDP percentiles and coverage.
This figure plots descriptive statistics for BDP. Panel A shows cross-sectional percentiles of BDP over the sample. Panel B plots the monthly coverage of BDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.55	0.64	0.65	0.81	0.34
	[2.60]	[3.04]	[3.73]	[3.78]	[4.00]	[3.58]
α_{CAPM}	-0.11	-0.02	0.11	0.11	0.15	0.26
	[-1.64]	[-0.27]	[1.54]	[1.68]	[2.59]	[2.81]
α_{FF3}	-0.16	-0.10	0.03	0.05	0.14	0.30
	[-2.65]	[-1.55]	[0.52]	[0.92]	[2.34]	[3.31]
α_{FF4}	-0.18	-0.12	0.00	0.05	0.20	0.38
	[-2.91]	[-1.89]	[0.00]	[0.91]	[3.39]	[4.18]
α_{FF5}	-0.25	-0.23	-0.14	-0.07	0.13	0.37
	[-3.98]	[-3.71]	[-2.43]	[-1.40]	[2.09]	[4.01]
α_{FF6}	-0.25	-0.24	-0.15	-0.06	0.18	0.43
	[-4.07]	[-3.82]	[-2.65]	[-1.21]	[2.99]	[4.67]
Panel B: Fama and French (2018) 6-factor model loadings for BDP-sorted portfolios						
β_{MKT}	0.98	0.99	0.95	0.96	1.06	0.08
	[66.75]	[67.58]	[71.59]	[80.18]	[75.23]	[3.76]
β_{SMB}	0.06	0.03	-0.08	-0.17	-0.03	-0.09
	[2.87]	[1.54]	[-4.16]	[-9.39]	[-1.20]	[-2.71]
β_{HML}	0.07	0.12	0.10	0.09	0.01	-0.07
	[2.65]	[4.17]	[3.93]	[3.90]	[0.26]	[-1.61]
β_{RMW}	0.14	0.26	0.34	0.26	0.06	-0.08
	[4.85]	[9.04]	[13.21]	[11.10]	[2.23]	[-1.82]
β_{CMA}	0.16	0.19	0.22	0.14	0.01	-0.15
	[3.75]	[4.43]	[5.83]	[4.13]	[0.14]	[-2.43]
β_{UMD}	0.01	0.01	0.02	-0.01	-0.08	-0.09
	[0.87]	[1.00]	[1.52]	[-1.07]	[-5.50]	[-4.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	355	344	315	282	297	
me (\$$10^6$)	916	1093	1946	2407	2058	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the BDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34	0.26	0.30	0.38	0.37	0.43
			[3.58]	[2.81]	[3.31]	[4.18]	[4.01]	[4.67]
Quintile	NYSE	EW	0.22	0.18	0.16	0.20	0.15	0.18
			[3.79]	[3.11]	[2.82]	[3.46]	[2.60]	[3.11]
Quintile	Name	VW	0.36	0.30	0.34	0.39	0.41	0.44
			[3.91]	[3.24]	[3.72]	[4.17]	[4.37]	[4.69]
Quintile	Cap	VW	0.32	0.23	0.31	0.39	0.41	0.47
			[3.13]	[2.31]	[3.23]	[4.00]	[4.20]	[4.77]
Decile	NYSE	VW	0.51	0.41	0.47	0.54	0.55	0.61
			[4.02]	[3.33]	[3.81]	[4.31]	[4.39]	[4.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.31	0.22	0.26	0.31	0.31	0.35
			[3.25]	[2.43]	[2.84]	[3.41]	[3.37]	[3.80]
Quintile	NYSE	EW	0.10	0.07	0.05	0.07	0.02	0.04
			[1.67]	[1.19]	[0.80]	[1.24]	[0.36]	[0.76]
Quintile	Name	VW	0.33	0.26	0.30	0.33	0.35	0.37
			[3.56]	[2.86]	[3.25]	[3.57]	[3.76]	[3.98]
Quintile	Cap	VW	0.29	0.19	0.26	0.31	0.34	0.37
			[2.86]	[1.92]	[2.71]	[3.21]	[3.49]	[3.85]
Decile	NYSE	VW	0.47	0.36	0.41	0.45	0.47	0.50
			[3.69]	[2.91]	[3.32]	[3.67]	[3.76]	[4.01]

Table 3: Conditional sort on size and BDP

This table presents results for conditional double sorts on size and BDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDP and short stocks with low BDP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

		BDP Quintiles					BDP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.79	1.03	0.90	0.75	0.96	0.16	0.15	0.14	0.13	0.17	0.15
		[3.58]	[3.12]	[4.02]	[3.55]	[4.17]	[1.79]	[1.64]	[1.52]	[1.36]	[1.76]	[1.58]
	(2)	0.61	0.66	0.81	0.91	0.86	0.25	0.25	0.18	0.21	0.13	0.16
		[2.70]	[2.90]	[3.60]	[4.24]	[3.74]	[2.64]	[2.63]	[1.95]	[2.23]	[1.32]	[1.62]
	(3)	0.63	0.72	0.81	0.82	0.88	0.26	0.23	0.20	0.27	0.20	0.25
		[2.92]	[3.42]	[3.89]	[4.00]	[4.00]	[2.84]	[2.56]	[2.23]	[2.89]	[2.17]	[2.69]
	(4)	0.63	0.63	0.68	0.75	0.90	0.27	0.23	0.21	0.26	0.24	0.27
		[3.13]	[3.24]	[3.39]	[3.83]	[4.21]	[3.17]	[2.74]	[2.52]	[3.06]	[2.75]	[3.16]
	(5)	0.42	0.58	0.57	0.65	0.78	0.36	0.26	0.32	0.40	0.40	0.46
		[2.34]	[3.34]	[3.27]	[3.72]	[3.76]	[3.30]	[2.45]	[3.02]	[3.74]	[3.75]	[4.29]

Panel B: Portfolio average number of firms and market capitalization

		BDP Quintiles					BDP Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	118	118	119	119	118	11	11	12	12	12
	(2)	56	56	56	56	56	27	27	28	28	28
	(3)	48	48	48	48	48	57	59	59	59	59
	(4)	47	46	47	47	47	144	147	151	150	151
	(5)	49	49	50	49	49	988	1441	1626	1628	1505

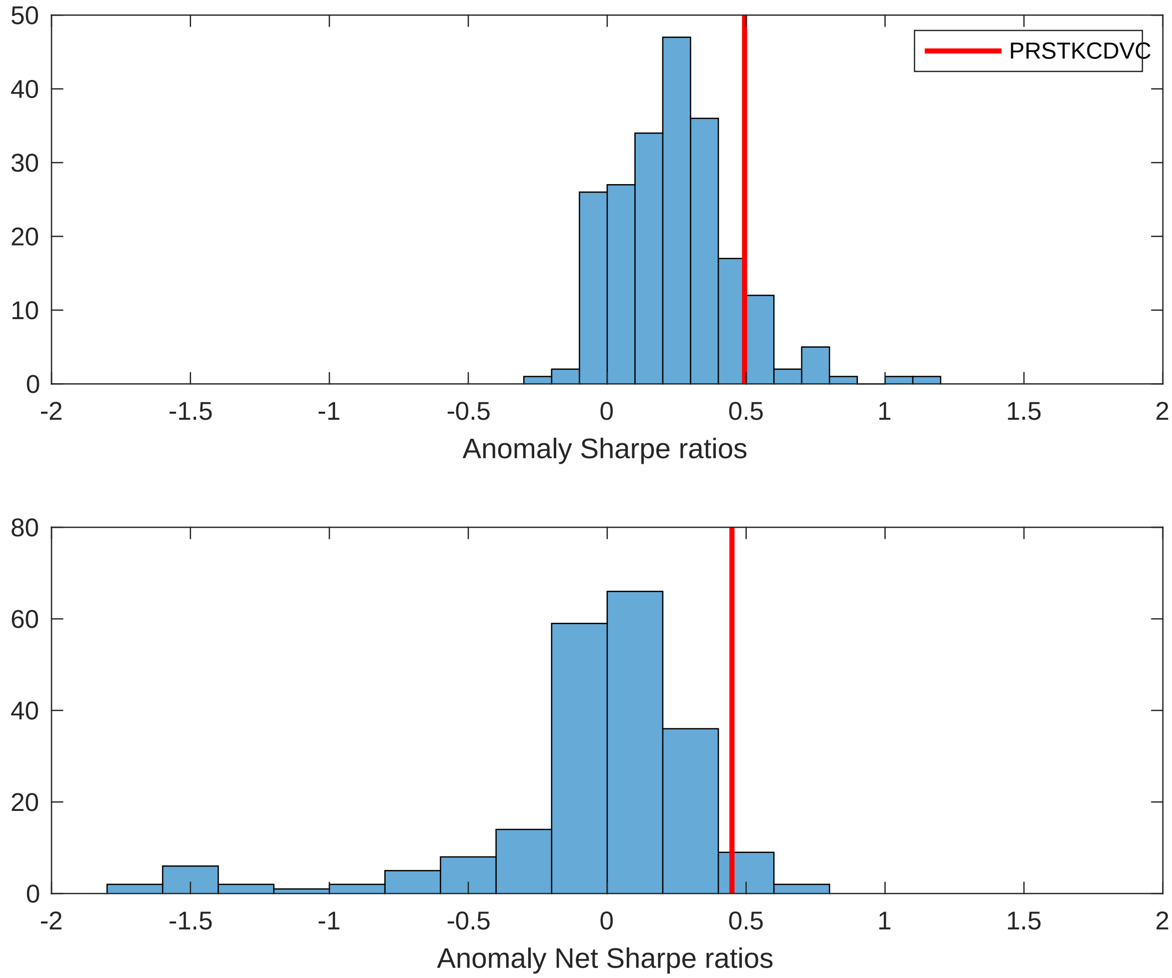


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

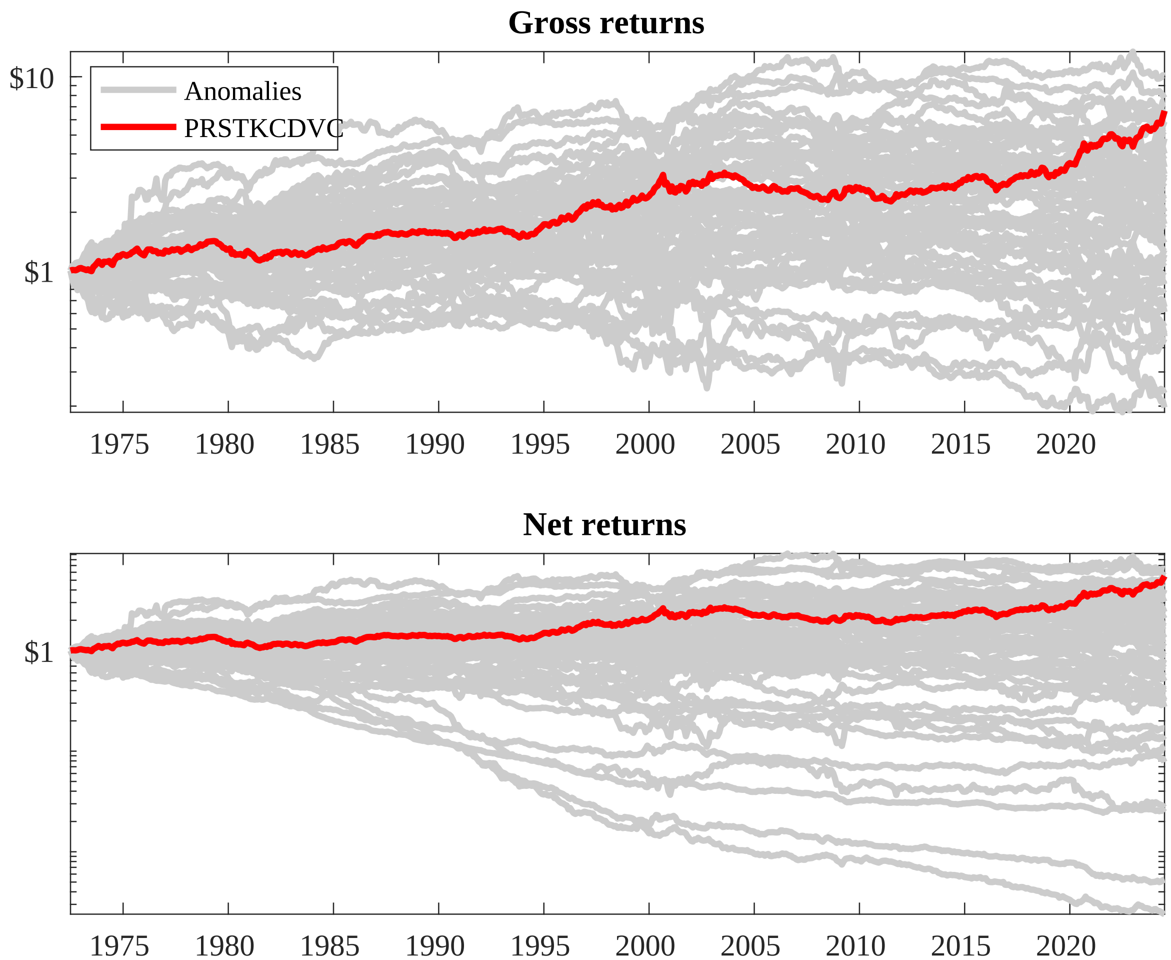
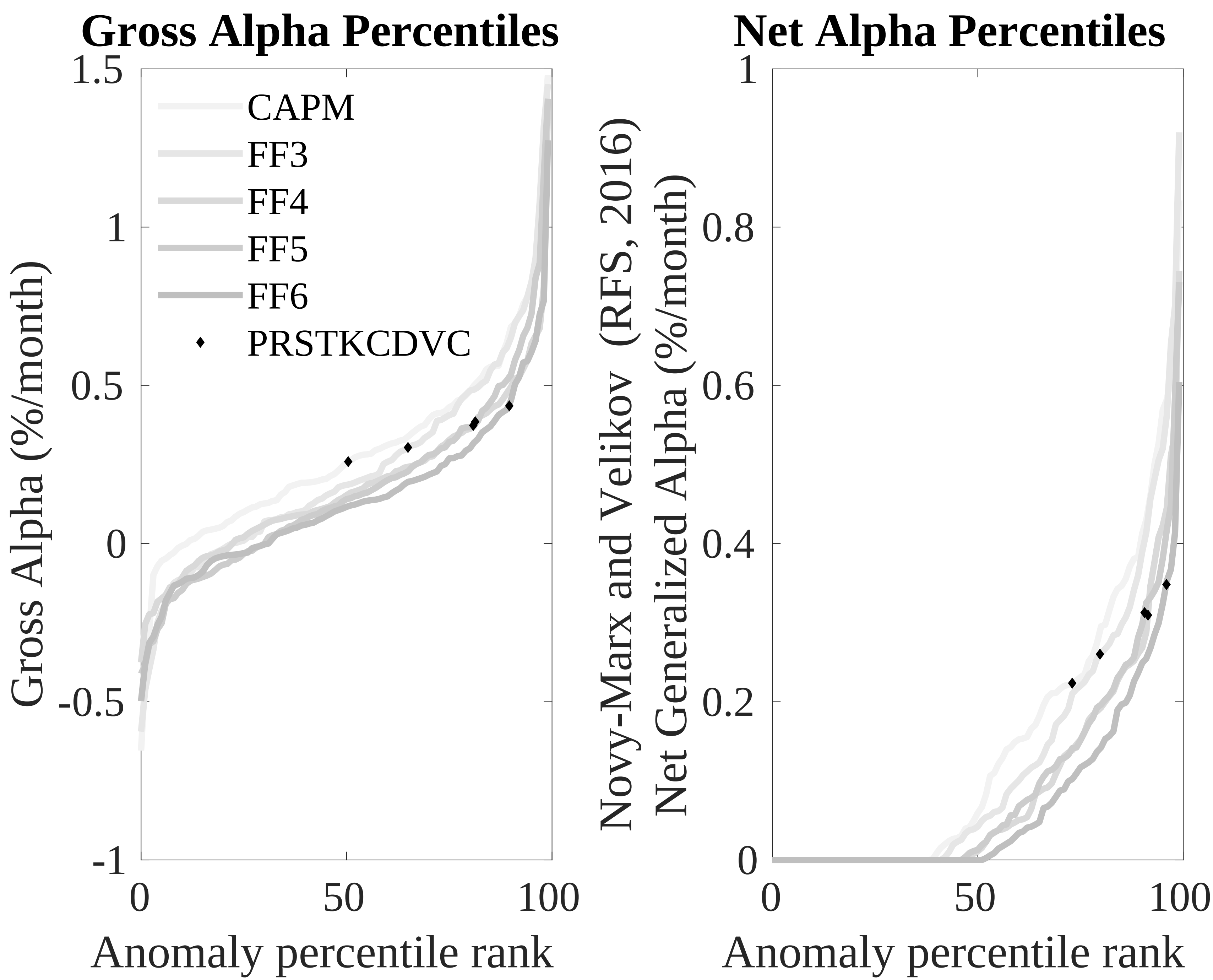


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
0
50
100
Gross Alpha (%/month)
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
PRSTKCDVC
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
0.8
0.6
0.4
0.2

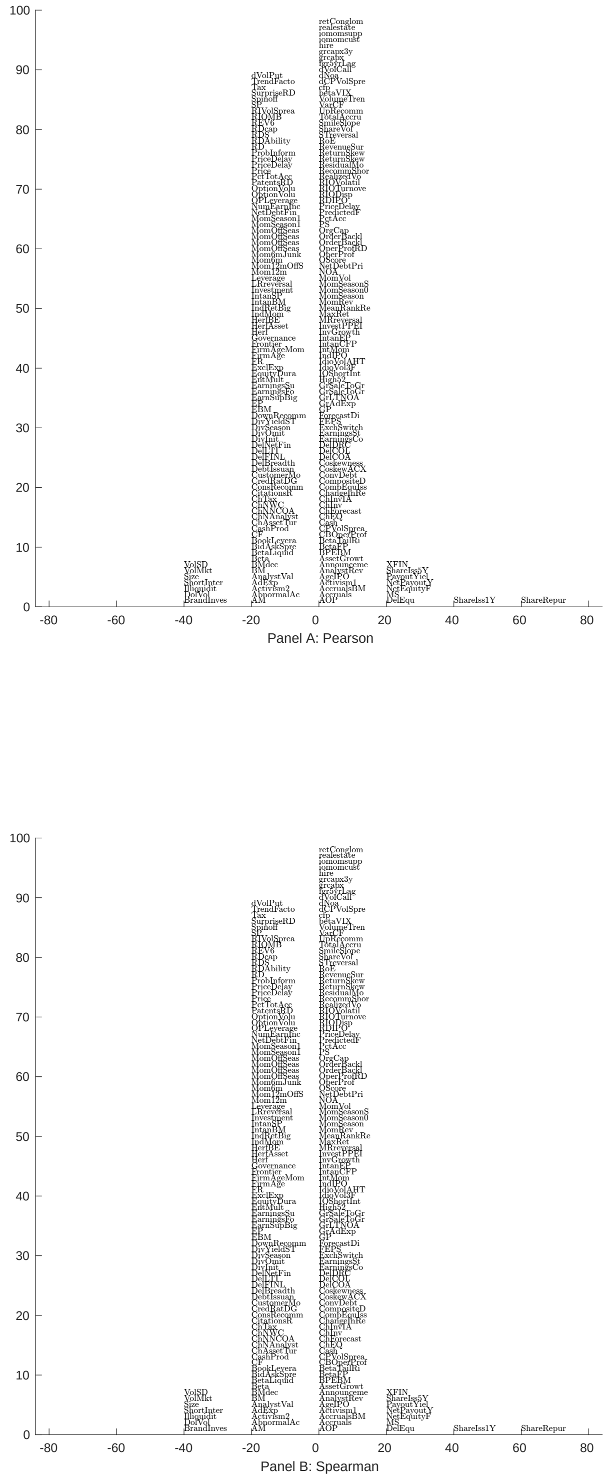


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 203 filtered anomaly signals with BDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

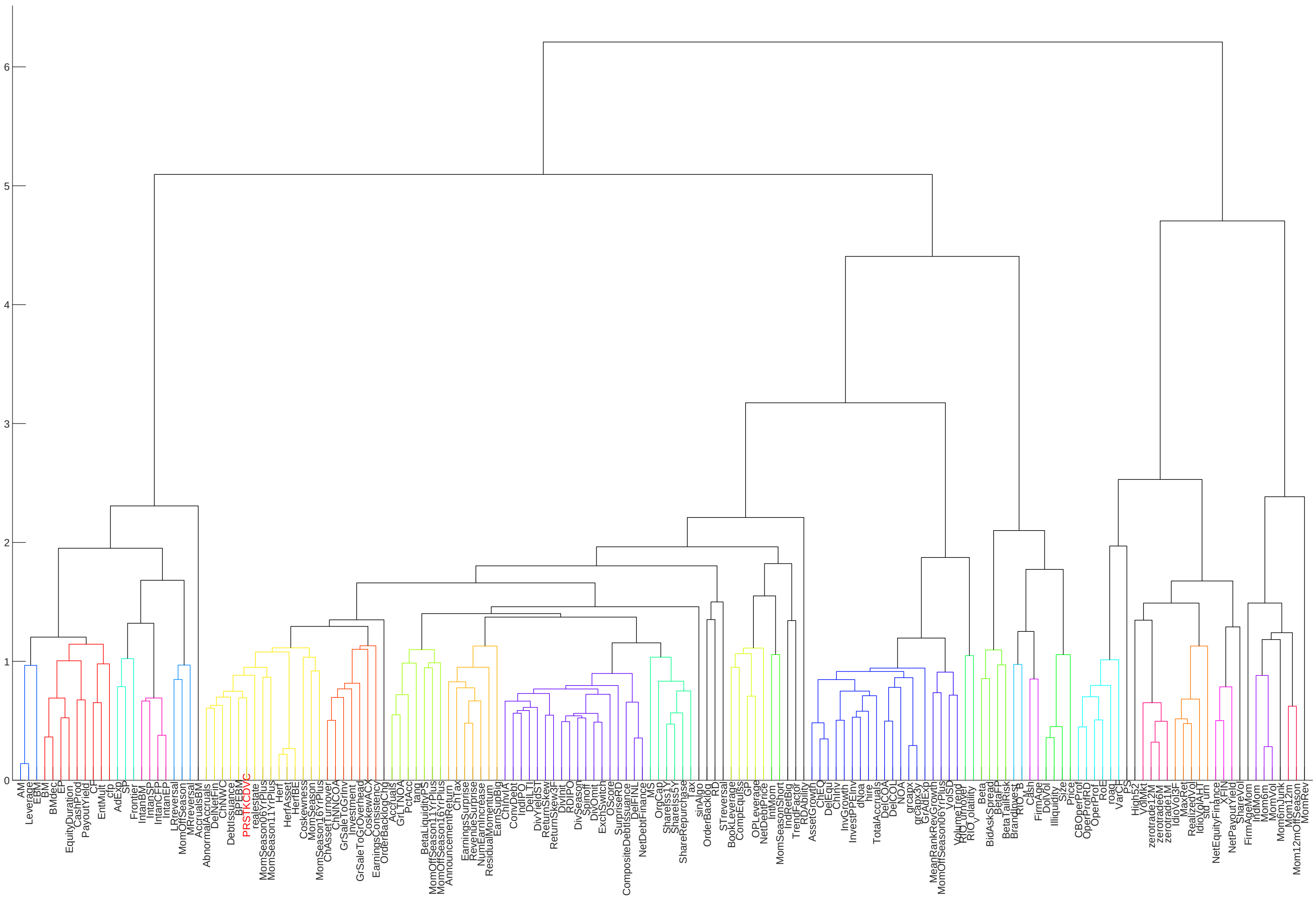


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

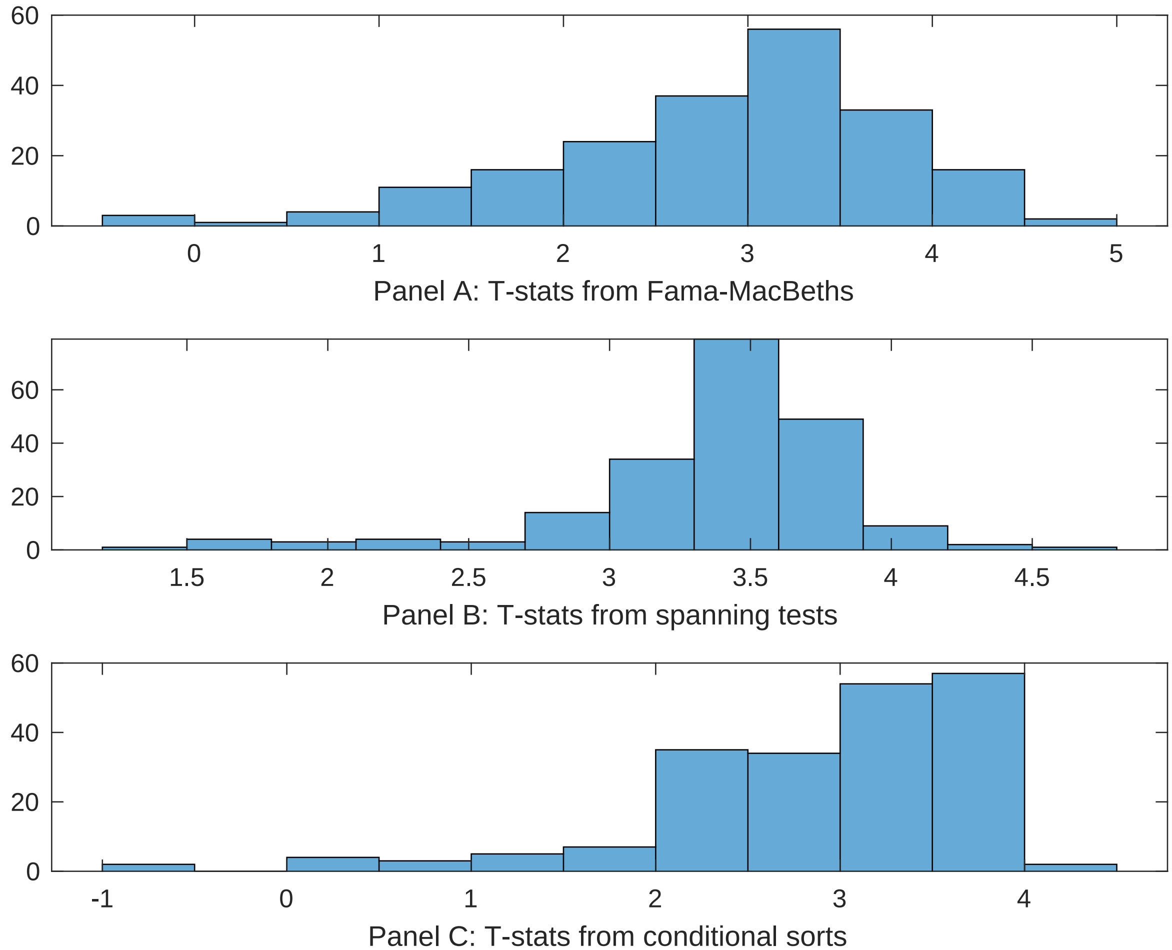


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP} BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDP} BDP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.11	0.12	0.11	0.12	0.11	0.11	0.11
	[5.48]	[7.42]	[5.48]	[6.14]	[5.54]	[4.93]	[6.06]
BDP	0.27	0.33	0.26	0.30	0.26	0.28	0.29
	[2.76]	[3.33]	[2.64]	[3.07]	[2.61]	[3.03]	[3.18]
Anomaly 1	0.32						-0.28
	[1.12]						[-0.37]
Anomaly 2		0.28					0.34
		[2.20]					[0.25]
Anomaly 3			0.39				0.22
			[0.05]				[1.32]
Anomaly 4				0.30			0.25
				[2.31]			[2.31]
Anomaly 5					0.16		0.20
					[0.30]		[0.95]
Anomaly 6						0.17	0.45
						[1.03]	[0.31]
# months	619	619	619	619	619	614	614
$\bar{R}^2(\%)$	1	2	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the BDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.38	0.34	0.40	0.42	0.38	0.35	0.28
	[4.26]	[3.77]	[4.46]	[4.63]	[4.25]	[3.90]	[3.05]
Anomaly 1	-17.02						-2.62
	[-6.16]						[-0.35]
Anomaly 2		-19.27					-24.00
		[-6.89]					[-4.01]
Anomaly 3			-15.61				9.14
			[-5.66]				[1.11]
Anomaly 4				-21.04			-1.32
				[-4.15]			[-0.22]
Anomaly 5					-15.93		2.32
					[-5.82]		[0.37]
Anomaly 6						-20.77	-19.37
						[-5.97]	[-5.07]
mkt	0.88	-0.96	1.33	3.81	1.11	3.57	-3.66
	[0.37]	[-0.39]	[0.56]	[1.62]	[0.46]	[1.62]	[-1.48]
smb	-8.05	-17.77	-9.83	4.71	-12.56	4.76	-6.18
	[-2.58]	[-5.17]	[-3.11]	[1.09]	[-3.87]	[1.28]	[-1.17]
hml	-1.47	0.05	-1.42	0.56	-1.44	-5.00	0.22
	[-0.36]	[0.01]	[-0.35]	[0.13]	[-0.35]	[-1.25]	[0.05]
rmw	2.83	3.48	2.90	-0.70	2.79	-3.75	4.22
	[0.64]	[0.80]	[0.64]	[-0.16]	[0.62]	[-0.91]	[0.94]
cma	-5.94	-6.30	-7.13	-13.38	-7.64	-6.65	-1.03
	[-0.96]	[-1.03]	[-1.15]	[-2.17]	[-1.23]	[-1.07]	[-0.17]
umd	-7.76	-4.05	-8.16	-8.36	-8.61	-7.31	-1.55
	[-3.75]	[-1.87]	[-3.94]	[-3.98]	[-4.16]	[-3.49]	[-0.63]
# months	620	620	620	620	620	615	615
$\bar{R}^2(\%)$	17	18	16	14	16	16	21

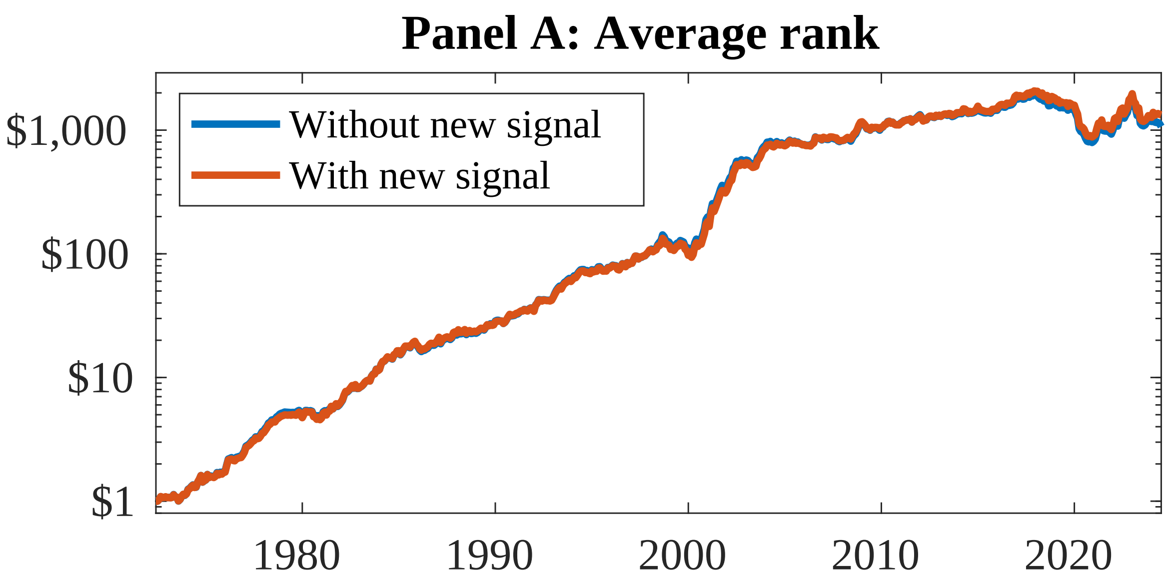


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cohen, L. and Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011.

DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Grullon, G. and Michaely, R. (2002). Dividends, share repurchases, and the substitution hypothesis. *Journal of Finance*, 57(4):1649–1684.

Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance*, 64(5):2289–2325.

Hong, H. and Stein, J. C. (1999a). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.

Hong, H. and Stein, J. C. (1999b). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.

Hou, K. (2007). Industry information diffusion and the lead-lag effect in stock returns. *Review of Financial Studies*, 20(4):1113–1138.

Ikenberry, D., Lakonishok, J., and Vermaelen, T. (1995). Market underreaction to open market share repurchases. *Journal of Financial Economics*, 39(2-3):181–208.

McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper.*

Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper.*